

Mid-term report
Dashboard and Storyboard on
Effect of Historical Redlining on Health Outcomes
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Introduction

In the 1930s, the Home Owners' Loan Corporation (HOLC) graded urban neighborhoods from “A” (least risky) to “D” (“hazardous,” commonly redlined) largely on the basis of racial and ethnic composition [1]. These grades shaped decades of mortgage lending, investment, and infrastructure decisions, and their legacy is thought to persist in the built environment and socioeconomic conditions of these neighborhoods today [1].

This project aimed to visually demonstrate how this structural inequality contributes to differences in obesity, diabetes, mental health, and overall health access across communities, using an interactive dashboard and narrative storyboard to communicate these patterns to a general audience.

Data source

We are using multiple data for this project which included:

- a. HOLC historical redlining maps, digitized HOLC neighborhood boundaries and grades (A/B/C/D) for graded U.S. cities [1].
- b. PLACES: Census Tract Data , model-based small-area estimates of health measures (e.g., obesity, diabetes, poor mental health days, access to health insurance/care) at the census tract level [2].
- c. CDC Social Vulnerability Index, tract-level composite index of socioeconomic status, household composition, housing/transportation, and minority status used as a control/context variable [3].

Study area

For this study, we are using census tract–level data for census tracts across the United States. Because HOLC “Security Maps” were only produced for a subset of U.S. cities, the redlining analysis is restricted to census tracts that fall within these historically graded areas; tracts outside graded areas are excluded from the treatment/comparison framework, though national PLACES and SVI data are used to provide broader context.

Methodology

Data Integration

HOLC redlining polygons, PLACES tract-level health estimates, and SVI tract data are being integrated into ArcGIS Pro using census tract id as the common join key. Because HOLC boundaries do not align with modern census tract boundaries, a spatial join/overlay approach is used to assign each historical HOLC grade to the census tract(s) it overlaps.

Each census tract was assigned the HOLC grade using area-weighted average grade was calculated where a tract overlapped multiple HOLC zones. The tracts with no HOLC overlap were coded as a separate 'ungraded' category.

Analytical Approach

The analysis compares health outcome measures (obesity, diabetes, poor mental health, and health care access) across HOLC grade categories (A, B, C, D), while accounting for social vulnerability as a contextual variable.

Software and Tools

Data cleaning was done in R version 4.2.1. Data were imported into ArcGIS pro for spatial joins, and exploration mapping and analysis. Finalized layers are published to ArcGIS Online and used to build an interactive ArcGIS Dashboard for exploring health outcomes by geography and HOLC grade, alongside an ArcGIS StoryMap that provides narrative context and guides the viewer through the findings.

Preliminary Results

The recent version of the dashboard is available in the link:

<https://www.arcgis.com/apps/dashboards/fd1934aafbca403596be88245995be9e>

The recent version of storyboard is available in the link:

<https://storymaps.arcgis.com/stories/60dd7049052f4366a497c6a77295bfbfd>

Challenges Encountered

We encountered following challenges during this project:

- Slow online ArcGIS pro makes it difficult to run the software.
- Difficulty in joining multiple datasets
- Missing or incomplete PLACES estimates for some tracts, particularly in smaller or rural areas.

Remaining Work and Timeline

The following tasks remain before the final deliverable:

Task	Target Date	Status
Acquire and clean HOLC redlining shapefiles	Completed	Done
Download and process PLACES census tract data	Completed	Done
Download CDC SVI data	Completed	Done
Join HOLC grades to modern census tracts	In progress	Ongoing
Descriptive/statistical comparison across HOLC grades	July 10	Ongoing

Build ArcGIS Pro map layers	July 15	Ongoing
Build online dashboard (ArcGIS Dashboard)	July 20	Ongoing
Build StoryMap narrative	July 20	Ongoing
Final report and presentation	July 23	Ongoing

References

1. Mapping Inequality. <https://dsl.richmond.edu/panorama/redlining/>. Accessed 4 July 2026.
2. PLACES: Census Tract Data (GIS Friendly Format), 2025 release | Data | Centers for Disease Control and Prevention. https://data.cdc.gov/500-Cities-Places/PLACES-Census-Tract-Data-GIS-Friendly-Format-2025-/yjkw-uj5s/about_data. Accessed 4 July 2026.
3. CDC. Social Vulnerability Index. Place and Health - Geospatial Research, Analysis, and Services Program (GRASP). 2024. <https://www.atsdr.cdc.gov/place-health/php/svi/index.html>. Accessed 4 July 2026.